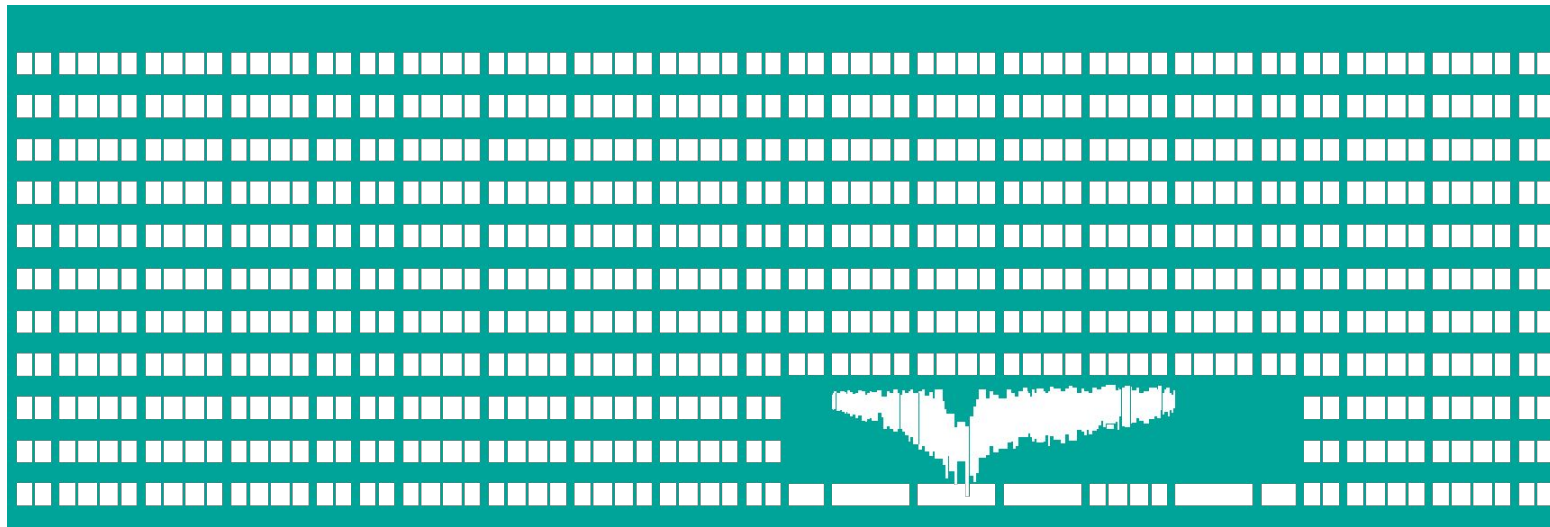
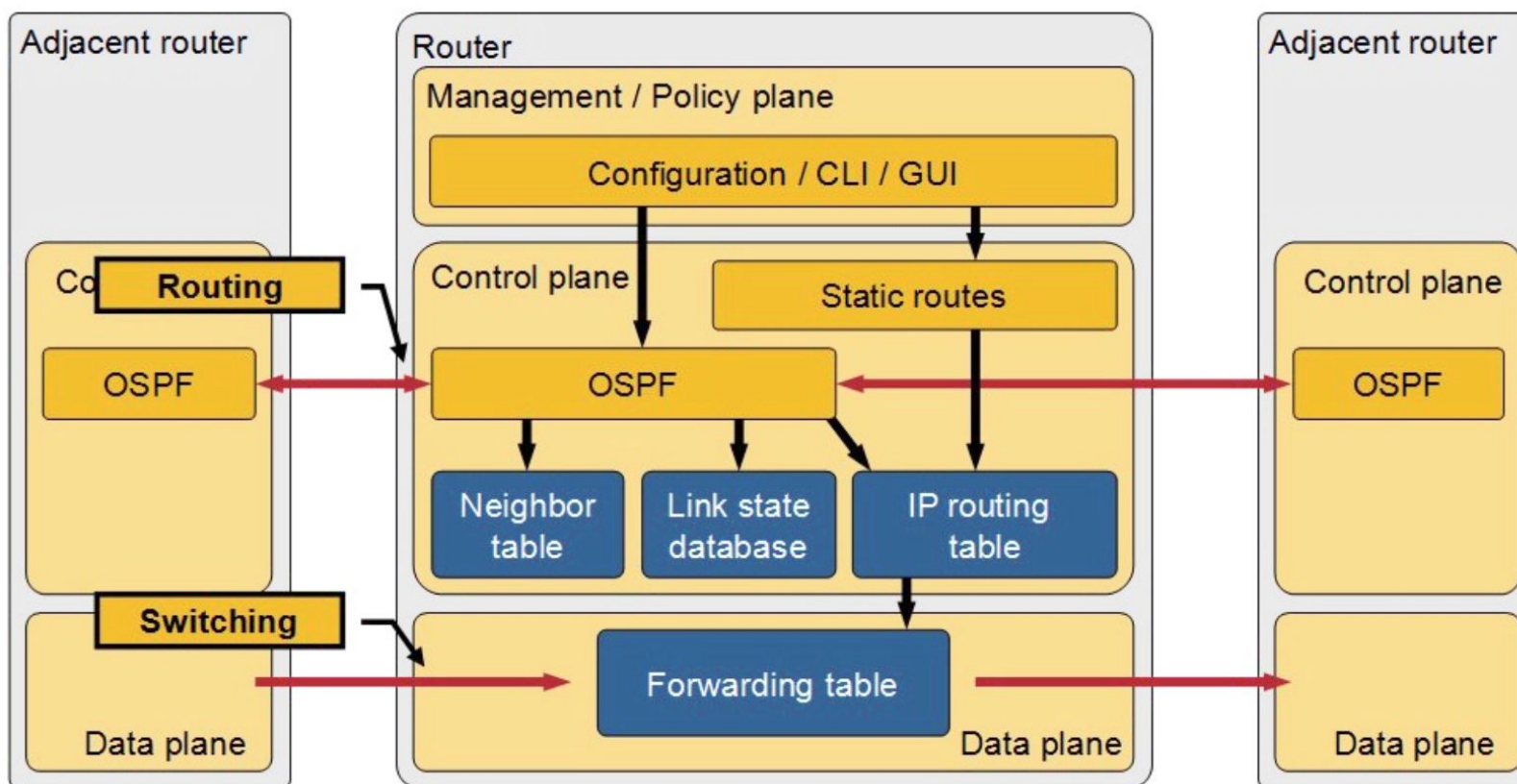




10 - SDN Basics with OpenFlow

Ing. Jan Rozhon, Ph.D.

Traditional Networks Overview



[\[source\]](#)

Limitations of Traditional Networks

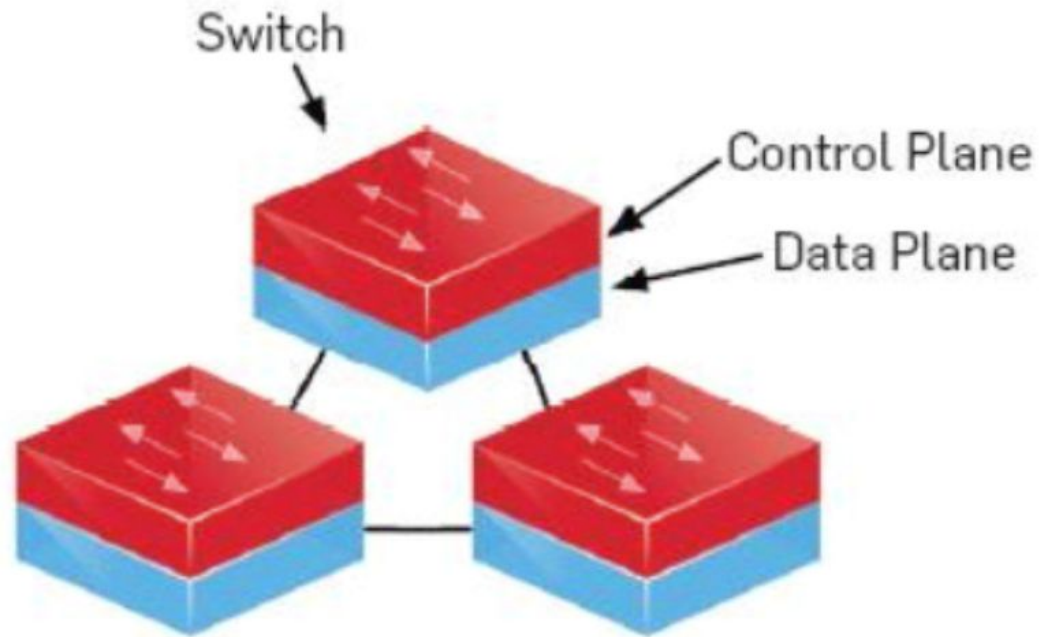
- Limited scaling
 - difficult maintenance of hundreds of network units.
- Vendor Lock-in
 - compatibility issues.
- Slow development
 - IPv6 vs Android App,
 - limited pool of developers.
- Complicated resource migration
 - Virtual machines migration,
 - Containers.
- Limited resource sharing
 - VLANs.

Idea of SDN

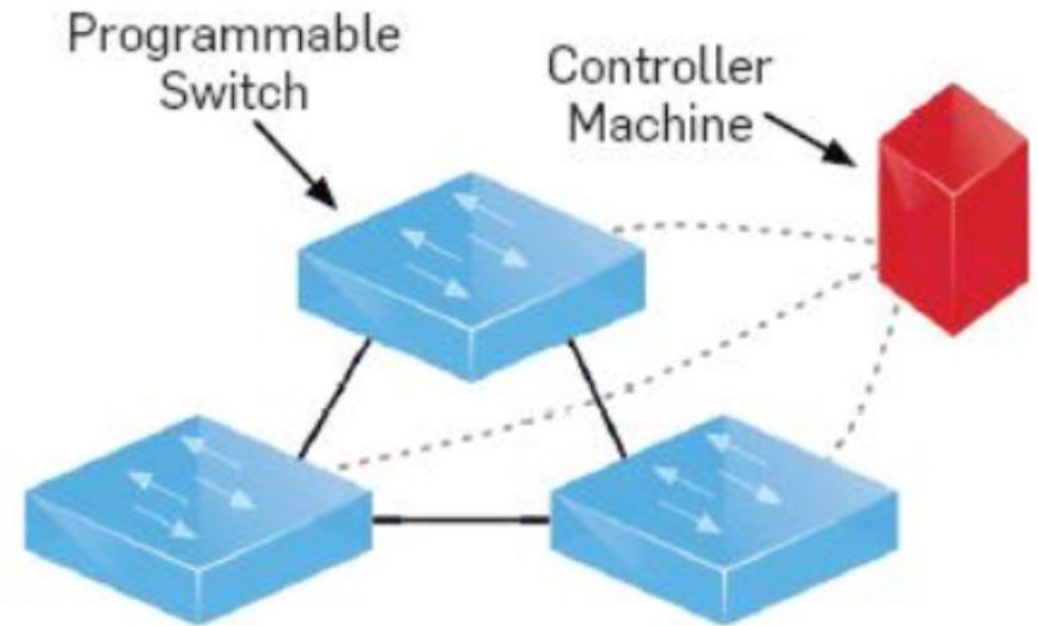
- Separation of the Control and Data Plane
 - network elements are simple,
 - there is one software application that controls the routing logic for all the elements - the controller,
 - Note: Remember the CSS signaling systems?
- Idea already present in the past:
 - FORCES (2003)
 - new standardization,
 - changes in hardware required.
 - RCP (Routing Control Platform, 2004)
 - limited by the common protocols.

Idea of SDN

Traditional Network



Software-Defined Network



Challenges

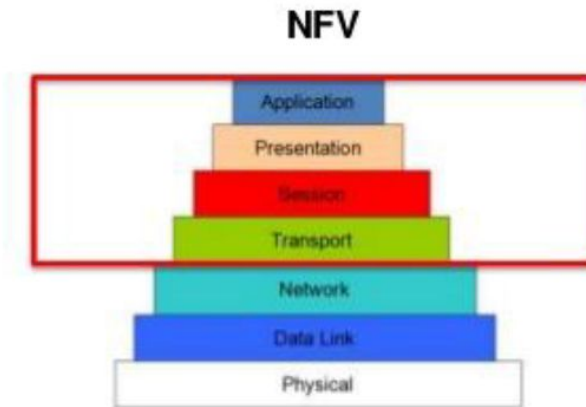
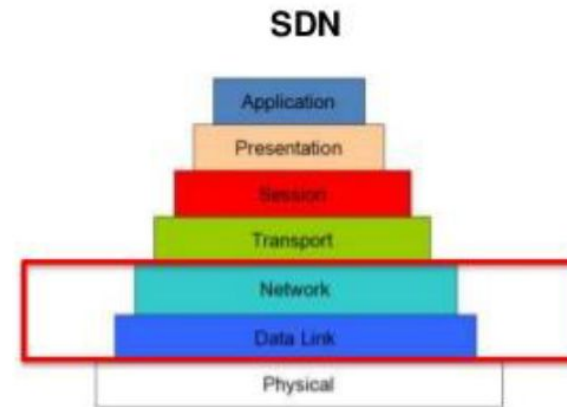
- Scalability
 - eliminate redundancy,
 - routing tables,
 - routing computations.
 - accelerated lookups,
 - maintain indices of affected routers.
 - network partitioning (use of IGP).
- Reliability
 - hot spare,
- Consistency
 - each controller needs to have its own information feed.

Advantages

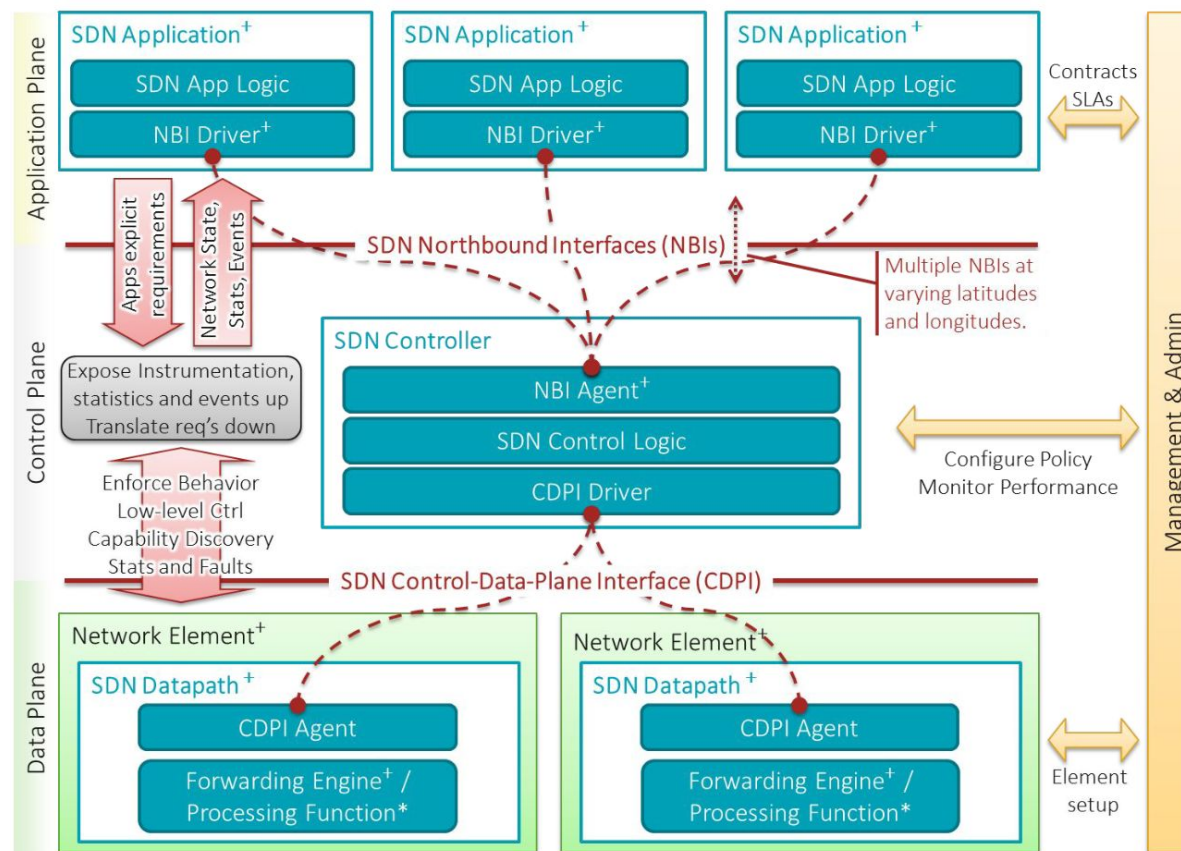
- Virtualization
- Resource Sharing
 - isolation on the level of CPU/application,
 - possible coexistence of production and testing environment.
- Vendor Lock-in prevention
- Rapid Development
- Continuous Development
 - the control logic can evolve in time after you buy switch/router.
- Dynamic Resource Allocation
 - change in demand immediately reflected on the resources.
- Complete overview of the network partition
 - easier debugging.
- Lower costs
 - simpler hardware,
 - easier orchestration.

SDN vs NFV

- Network Function Virtualization
 - monolithic middleboxes (firewalls, load balancers...) replaced by modular software running on the
 - server,
 - example: Snort = IP Defragmenter + Preprocessing + Misuse Detection Engine + Logger,
 - allows for custom middlebox functions.



Architecture (one of many)

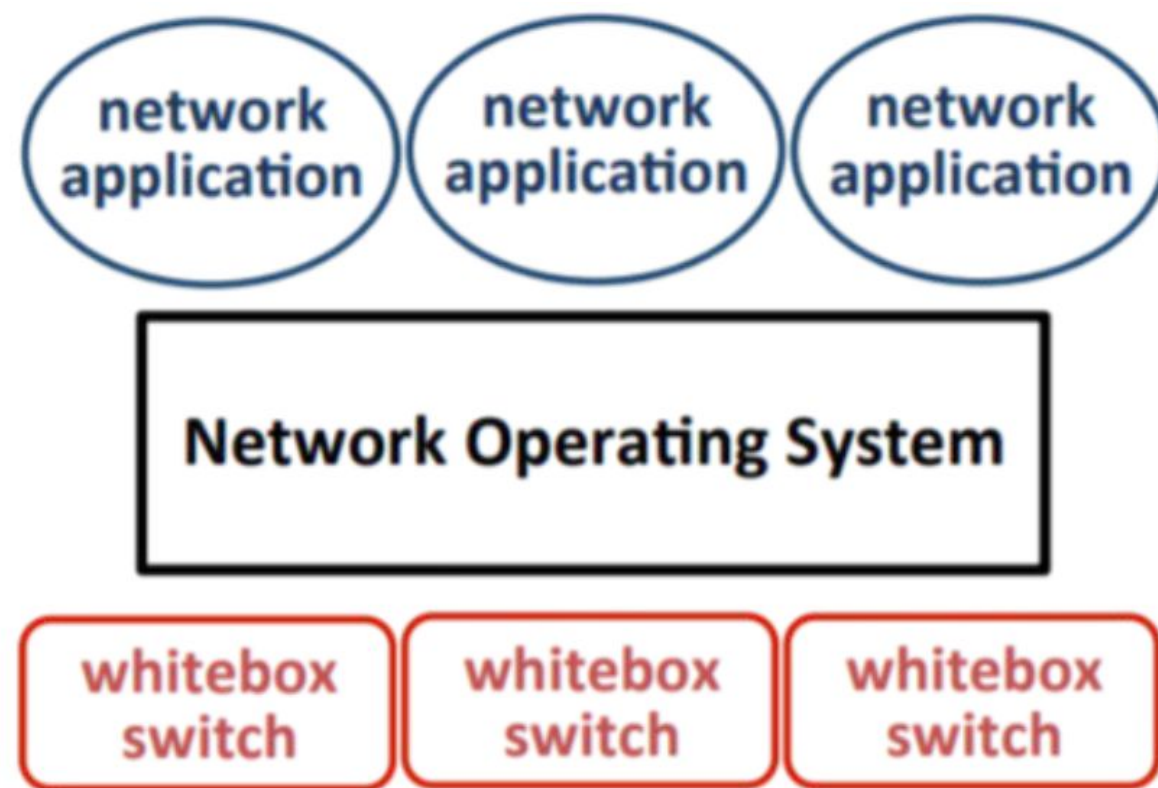
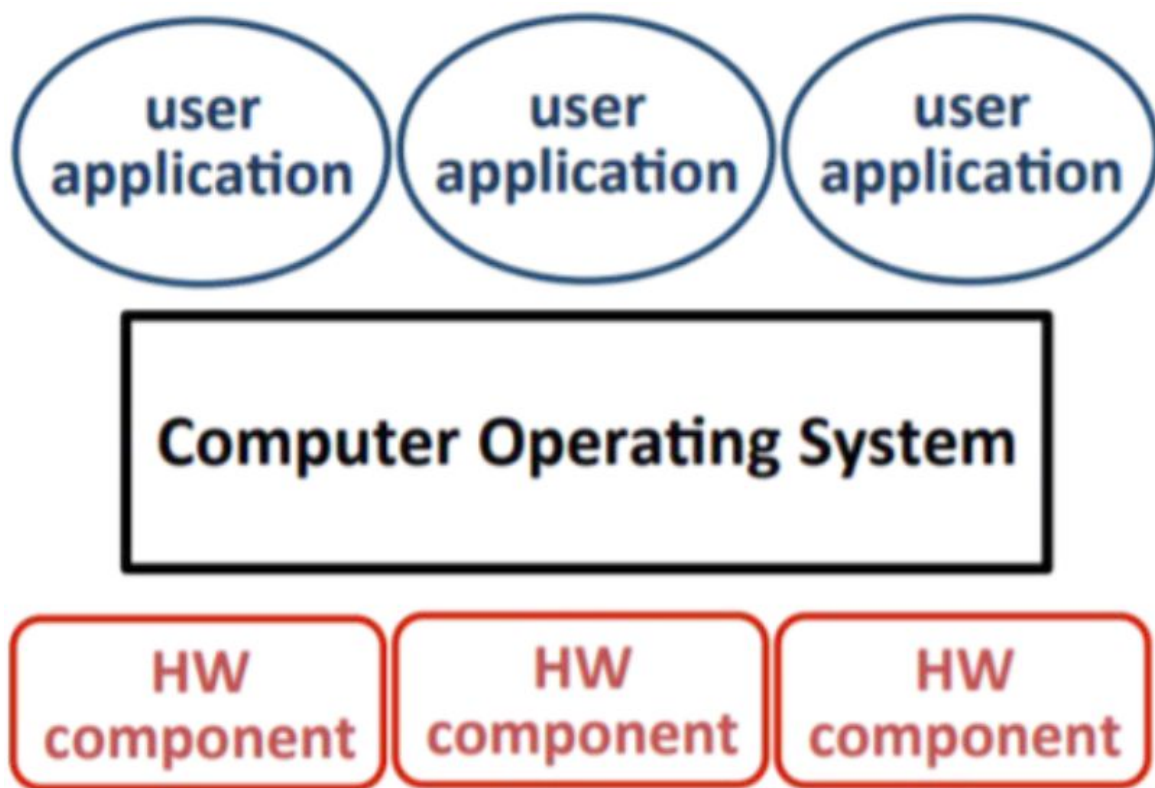


⁺ indicates one or more instances | ^{*} indicates zero or more instances

SDN Interfaces

- Northbound Interface
 - interface between application logic and a controller,
 - not standardized,
 - commonly REST.
- Southbound Interface
 - interface between a controller and the network elements,
 - standardized,
 - **still** common protocol - OpenFlow.
- SDNi
 - Inter SDN Controller Communication Interface,
 - interface between the controllers.

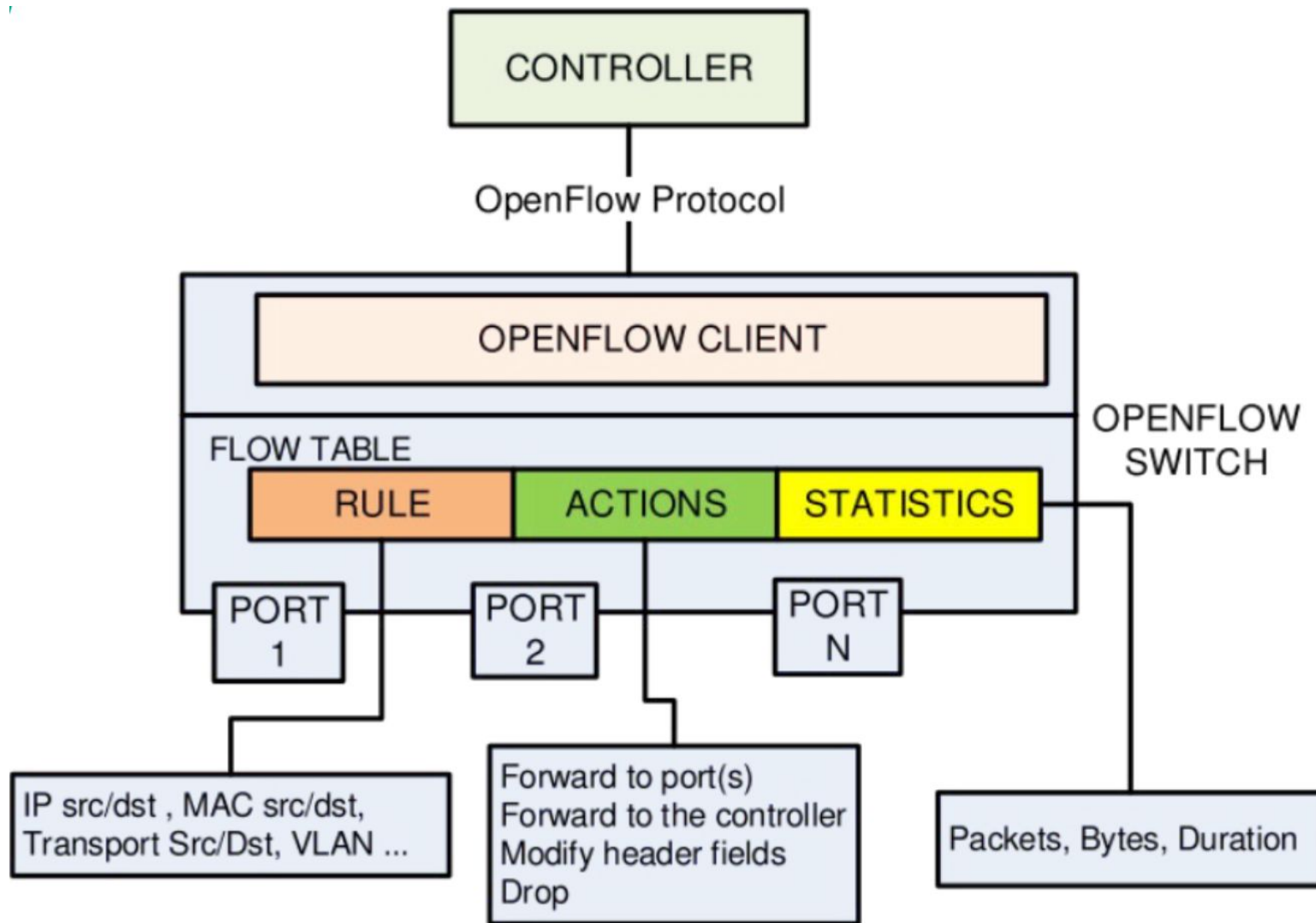
Architecture - The Concept



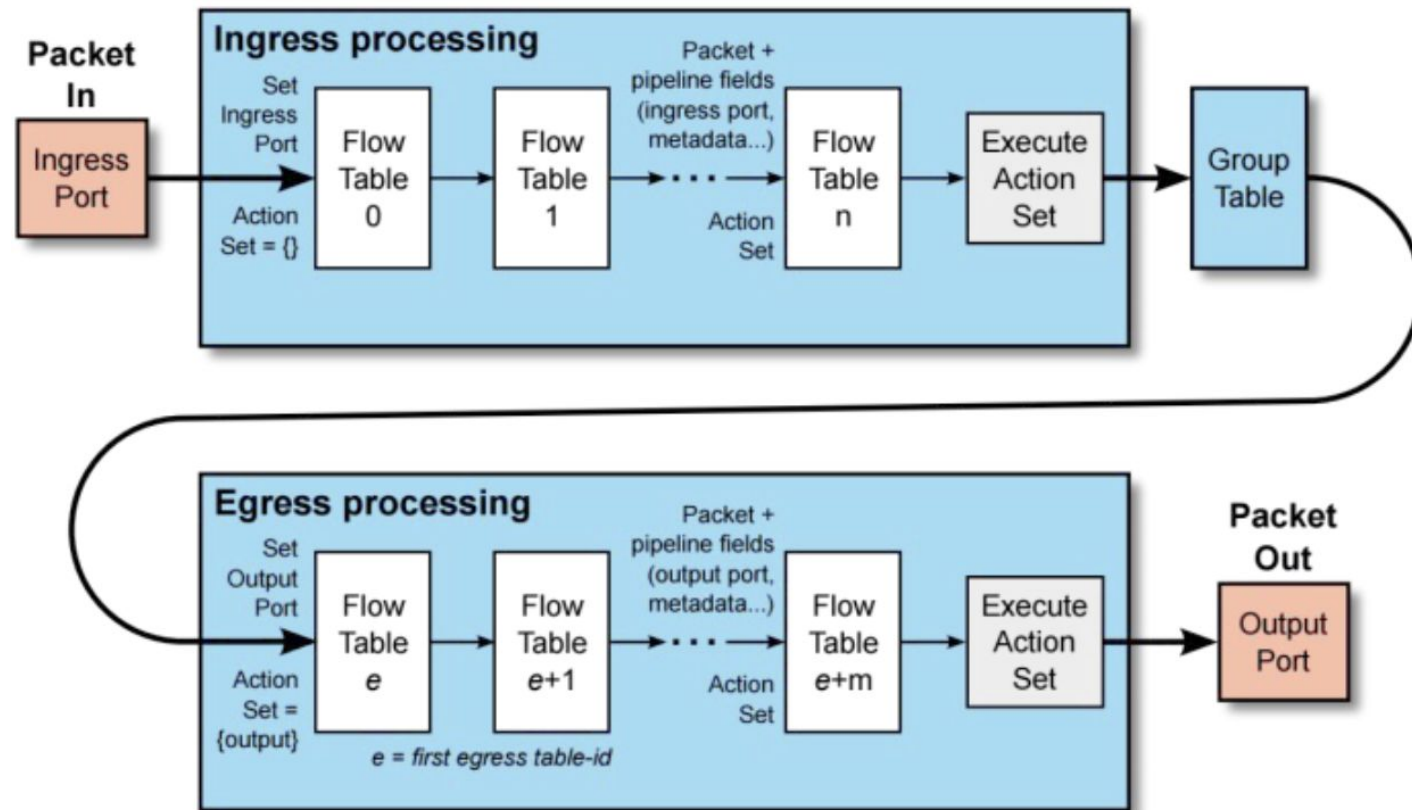
OpenFlow

- First defined in 2008 by Open Networking Foundation (ONF)
- Protocol to communicate between network element and separate controller in order to install forwarding
- rules into the element's flow table
- Implemented on the Southbound Interface (SDPI)
- Does not require custom hardware
 - flow tables common in 2008 hardware,
 - only needs to open an interface to separate controller,
 - hardware can still be closed, there is only need for open communication interface,
- **Now discontinued development** (since 2018)
- **Still in active use**

OpenFlow



OpenFlow Switch - Processing Pipeline



OpenFlow

- Each OpenFlow switch contains at least 1 flow table and a set of flow entries within that table
- In flow entries, there are:
 - match fields,
 - priority,
 - counters,
 - actions to be applied,
 - timeouts for flow expiration.
- In flow tables, flow entries are prioritized,
- Flow tables are connected sequentially allowing for cascaded execution of actions,
- If no match is found, Table-miss flow entry is executed,
 - matches anything,
 - priority 0,
 - usually continue to the next flow table OR ask controller.

OpenFlow

- If present, a Group Table provides additional layer for packet processing
 - action buckets (up to 32),
 - can be referenced by the flow table entry.
- Since OpenFlow 1.3 meter table can also be present
 - rate limiting,
 - DSCP capable,
 - meter table entry contains:
 - meter identifier,
 - meter bands,
 - counters.

OpenFlow

- Physical ports
 - Hardware interface
- Logical Ports
 - For Link aggregation group, tunnel, loopback interfaces
 - Every packets comes from/destined logical port have Tunnel-ID
- Reserved Ports – used for forwarding actions, defines sending to the controller, flooding or forwarding using non-OpenFlow methods
 - ALL - flooding all except ingress,
 - CONTROLLER - send packet to controller,
 - NORMAL - traditional forwarding,
 - FLOOD - traditional flooding,
 - IN_PORT - send packet out through ingress.

OpenFlow - Communication

- OpenFlow messages carry
 - OF switch configuration requests
 - Requests from OF switch to controller
- Mostly TCP or TCP/TLS (port 6633)
 - Separate TCP/IP network between controller(s) and OF switches
 - Connection initiated by OF switch
 - Symmetric/Asynchronous/Controller-to-Switch messages
- In case of OF channel failure, OF switch may start to behave as standard switch

OpenFlow - Communication

- OF protocol defines three types of the OF messages:
 - Symmetric messages,
 - no solicitation, both directions,
 - Hello, Echo, Error.
 - Asynchronous messages
 - sent asynchronously based on an event,
 - switch-to-controller messages.
 - PacketIn, FlowRemoved, PortStatus
 - Controller-to-switch messages
 - initiated by the controller.
 - PacketOut, FlowMod, PortMod, FeatureReq/Res

Thank you for your attention

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